

Data Analyst Technical Assessment

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About This Report

This report presents the results of a technical assessment of a digital advertising campaign dataset. The objective of the analysis is to evaluate campaign performance, identify the factors associated with advertising effectiveness, and provide data-driven recommendations to improve Return on Advertising Spend (ROAS).

To accommodate both business stakeholders and technical reviewers, the report is organized into two complementary sections.

- **Main Report** — Presents the key business findings, interpretations, and recommendations in a concise, stakeholder-focused manner. Statistical results are summarized only where necessary to support business decisions.
- **Technical Appendices** — Provide the detailed analytical procedures supporting the main report, including the complete data quality assessment, data cleaning methodology, exploratory analyses, statistical assumption checks, hypothesis testing, and post-hoc analysis.

Readers interested primarily in the business implications may focus on the main report, while readers wishing to review the analytical methodology and supporting evidence may refer to the appendices.

This document also includes interactive cross-references. Figures, tables, equations, sections, and appendices references are hyperlinked throughout the report, allowing readers to navigate directly to the referenced content by clicking the corresponding reference.

Software Used

- Python 3.13.5 (data cleaning, statistical analysis, and data processing)
- Visual Studio Code (Python development environment)
- Microsoft Power BI (data visualization and dashboard development)
- TeXstudio (report preparation using L^AT_EX)

Task 1

Data Quality and Initial Exploratory Assessment

Before conducting the performance analysis, the advertising campaign dataset was assessed to ensure that it was suitable for business decision-making. A comprehensive data quality assessment was performed, including verification of data types, duplicate records, missing values, logical consistency, and potential outliers.

The complete details of these procedures are provided in Appendix A.

1.0.1 Data Cleaning Summary

The data cleaning process consist of:

- Verified variable data types
- Checked duplicate observations
- Removed observations(20 datapoints) with missing revenue.
- Retained structurally missing Video Completion Rate values.
- Retained valid view-through conversions.
- Removed observations(47 datapoints) with 0 clicks,impressions and purchases with non-zero purchase.
- Addressed severe **Spend** outliers.

Overall, the dataset was determined to be sufficiently complete and reliable for subsequent analysis.

1.0.2 Challenge

The primary business challenges include:

- Overall ROAS is substantially below the target value.
- Campaign performance varies considerably across advertising platforms, creative types, and product categories.

- Weekly ROAS has shown a declining trend, indicating decreasing advertising efficiency.
- Regional performance differences remain unclear.
- The impact of competitive events on campaign performance is uncertain.
- The effectiveness of different creative types and audience segments has not yet been established.
- Advertising spend must be reduced in underperforming areas while maintaining revenue growth.

1.0.3 Performance Metrics

We need numbers to evaluate campaign effectiveness. In that regard, four standard digital marketing performance metrics were considered.

Metric	Formula
CTR	Clicks / Impressions
CPC	Spend / Clicks
CVR	Purchases / Clicks
ROAS	Revenue / Spend

Table 1.1: Calculated performance metrics.

These metrics provide standardized measures of audience engagement, advertising efficiency, conversion effectiveness, and financial return.

1.1 Initial Exploratory Assessment

1.1.1 Business Overview

Following data cleaning, the overall Return on Advertising Spend (ROAS) is 0.96, which is below the company's target of 1.20. This indicates that the current advertising strategy is not generating sufficient revenue relative to advertising expenditure. Performance also varies across campaigns, platforms, creative types, product categories, and target audiences, suggesting opportunities to improve budget allocation and campaign effectiveness.

Since improving ROAS is the primary business objective, the subsequent analysis focuses on identifying the factors associated with both high- and low-performing advertising campaigns.

1.1.2 Key Performance Metrics

To evaluate advertising performance, the following baseline metrics were selected.

Metric	Business Relevance
Spend	Measures advertising investment and budget allocation.
Revenue	Measures the financial return generated by advertising activities.
ROAS	Primary indicator of advertising efficiency and campaign profitability.
CTR	Measures audience engagement with advertisements.
CPC	Measures the cost required to generate user engagement.
CVR	Measures the effectiveness of converting engaged users into customers.

Table 1.2: Baseline performance metrics used throughout the analysis.

Together, these metrics provide a comprehensive view of advertising performance, covering investment, engagement, conversion, and financial return.

1.1.3 Preliminary Insights

I created a power BI dashboard(file is inside github repository) to summarize the ROAS to gives us a main idea of the campaign ad ROAS across different categorical variables.

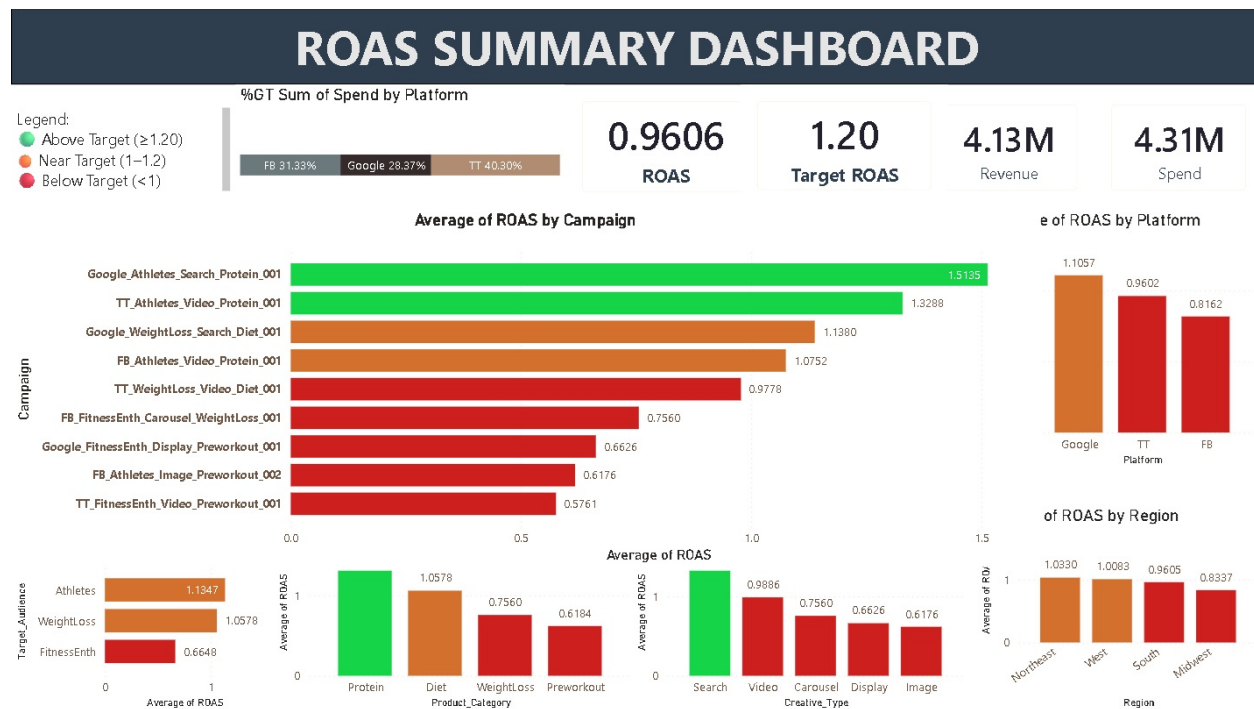


Figure 1.1: ROAS Dashboard Summary

The executive dashboard provides an initial overview of advertising performance prior to detailed analysis. Several observations can be made:

- Overall ROAS (0.96) remains below the business target of 1.20.

- Campaign performance varies substantially, with only a few campaigns exceeding the target ROAS.
- Google delivers the highest ROAS among advertising platforms, while Facebook records the lowest.
- Search creatives and Protein campaigns achieve the strongest advertising returns.
- Performance differences across regions are relatively small, whereas larger variation exists across creative types, product categories, and target audiences.

These observations establish a baseline understanding of current advertising performance and guide the subsequent analyses presented in Task 2.

Task 2

Performance Analysis

The analyses presented in this chapter should be interpreted as exploratory rather than causal. Campaign characteristics including platform, product_category, audience_target, creative_types, etc.. are not independently distributed across the dataset.

Consequently, observed performance differences cannot be attributed to any single factor in isolation. Instead, the objective of this chapter is to identify consistent performance patterns, prioritize areas for further investigation, and generate hypotheses to guide future optimization and controlled experiments.

However, descriptive statistics alone cannot determine whether these observed differences represent genuine performance differences or are simply the result of random variation. Therefore, statistical significance testing was performed before drawing business conclusions.

2.1 Statistical Methodology

The following statistical tests were used throughout the analysis:

1. Shapiro–Wilk Test

Used to assess whether the distribution of each performance metric follows a normal distribution. The results determined whether parametric statistical methods were appropriate for subsequent analyses.

Statistical Question:

- *Do the observed performance metrics satisfy the assumptions required for parametric statistical tests?*

2. Levene’s Test

Used to determine whether the variability (consistency) of a performance metric differs significantly across groups.

Business Question Answered:

- *Which business groups deliver more consistent advertising performance, and which exhibit greater variability or unpredictability?*

3. Kruskal–Wallis Test

A non-parametric alternative to one-way ANOVA used to determine whether the distribution of a performance metric differs significantly across multiple groups. This test was selected because the normality assumption required for ANOVA was not satisfied.

Business Question Answered:

- *Do the observed differences in campaign performance across business groups (e.g., platforms, regions, creative types, product categories, and target audiences) represent genuine differences, or are they likely due to random variation?*

4. Dunn's Post-hoc Test

Performed following a statistically significant Kruskal–Wallis test to identify which specific groups differ from one another while controlling for multiple comparisons.

Business Question Answered:

- *If significant differences exist, which specific business groups differ in their advertising performance?*

Together, these statistical procedures provide a systematic framework for validating assumptions, assessing performance consistency, determining whether differences are statistically significant, and identifying the specific groups responsible for those differences.

2.2 Channel Performance Analysis

Performance metrics across different channel/platforms was investigated and is shown in Fig. 2.1.

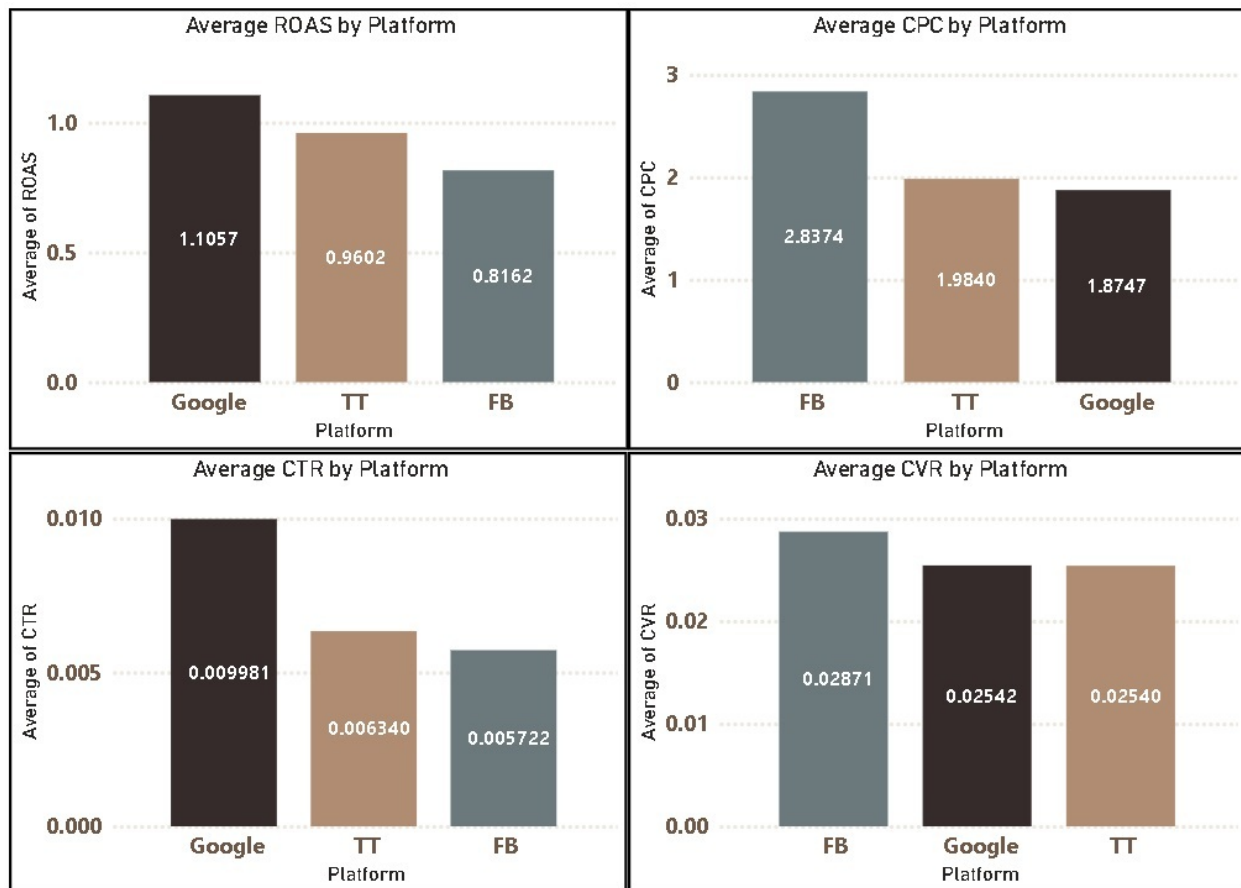


Figure 2.1: Performance Metrics by Platform

- FB exhibited the highest CVR out of all platforms which is offset by the high CPC and low CTR leading to its low ROAS.
- Google exhibited the highest CTR across platforms with the lowest Cost per Clicks.
- From a business perspective, Google currently delivers the greatest overall value in contrast to FB platform based on their ROAS.

Drilling down on platforms by creative.type we can see that Google search exhibited high ROAS while google Display exhibited one of the lowest ROAS as shown in Fig. 2.2.

This indicates that **the performance of an ad campaign is not tied to the platform**, further analyses is required.

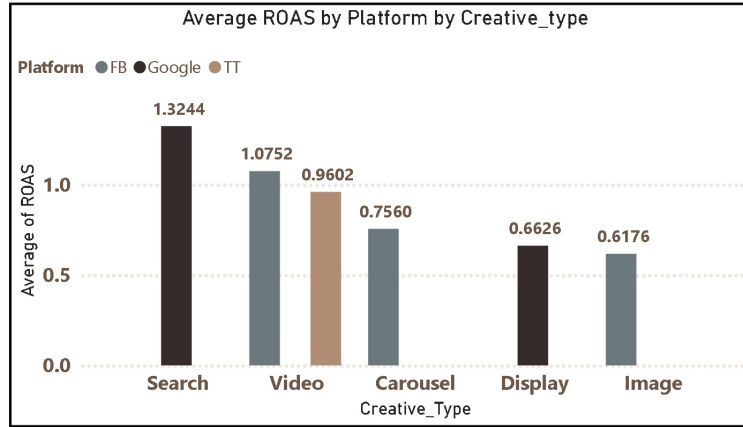


Figure 2.2: ROAS by Platform by Creative_type

2.3 Regional Performance Analysis

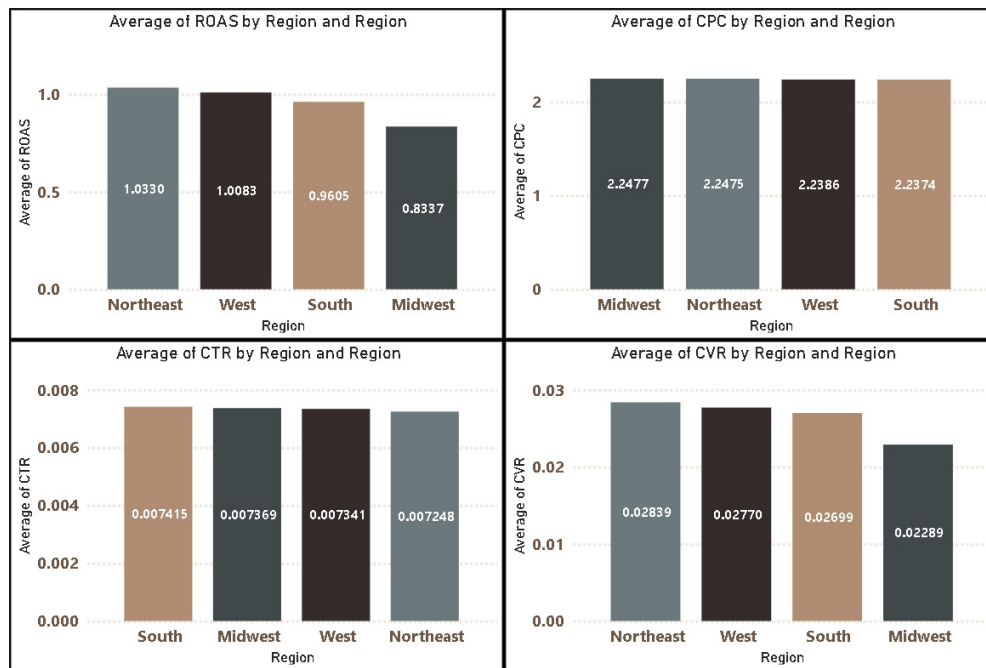


Figure 2.3: Performance Metrics by Region

- The **Midwest** consistently underperformed, recording both the lowest ROAS and the lowest conversion rate despite exhibiting click-through rates comparable to the other regions. This suggests that the primary challenge lies in converting interested users into customers rather than attracting engagement in this region.
- Because CTR and CPC did not differ significantly across regions, regional performance differences appear to be driven primarily by post-click behavior rather than differences in advertising cost or user engagement.

From a business perspective, further investigation should focus on **identifying factors that contribute to the Midwest's weaker conversion performance**.

Potential explanations include regional differences in customer preferences, purchasing behavior, market competition, or campaign targeting strategies.

2.4 Creative Performance Analysis

Creative formats are highly imbalanced across campaigns, with certain creative types represented by only one or two campaigns as shown in Fig. 2.4. Consequently, comparisons should be interpreted as descriptive rather than causal.

Creative_Type	Count of Campaign
Carousel	1
Display	1
Image	1
Search	2
Video	4
Total	9

Figure 2.4: Creative Type Campaign counts

Accordingly, the following observations are intended to describe aggregate performance patterns rather than isolate the causal effect of creative format.

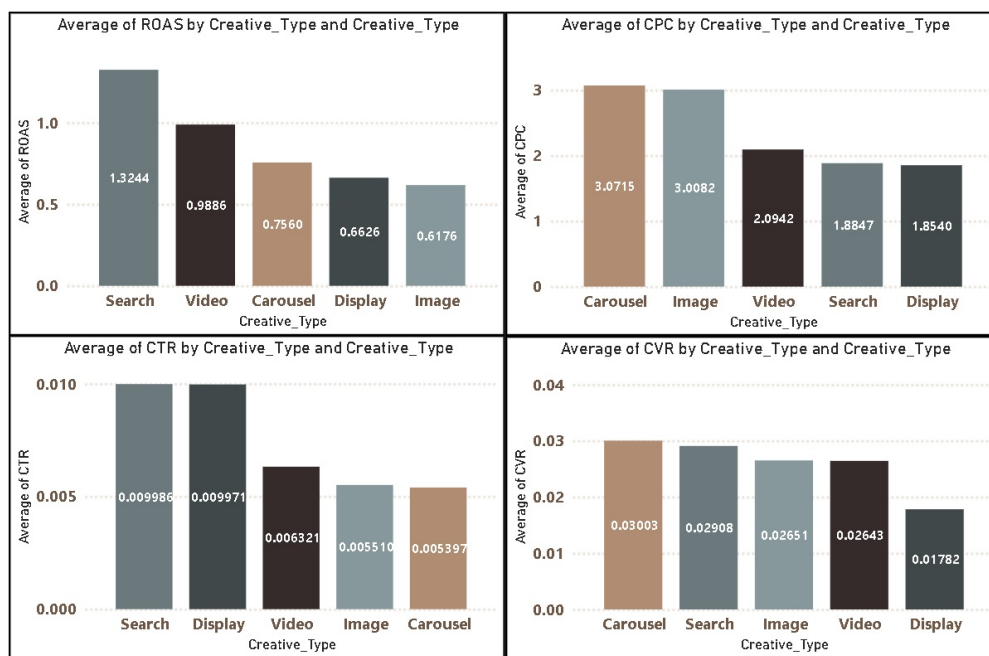


Figure 2.5: Performance Metrics by Creatives

Statistical significance tests for this section is shown in detail in Appendix B.3.

- Search creatives exhibited the highest ROAS, whereas Display and Image exhibited the lowest ROAS.
- Search and Display successfully generate clicks as shown by their CTR.
- In terms of traffic acquisition cost, carousel and image creatives are the most expensive.
- Carousel and Search exhibit the highest conversion efficiency as evidenced by their high CVR.

2.4.1 Video Completion Rate vs Conversion

We explored the relationship between video completion rate and conversion efficiency(CVR) as shown in Fig. 2.6.

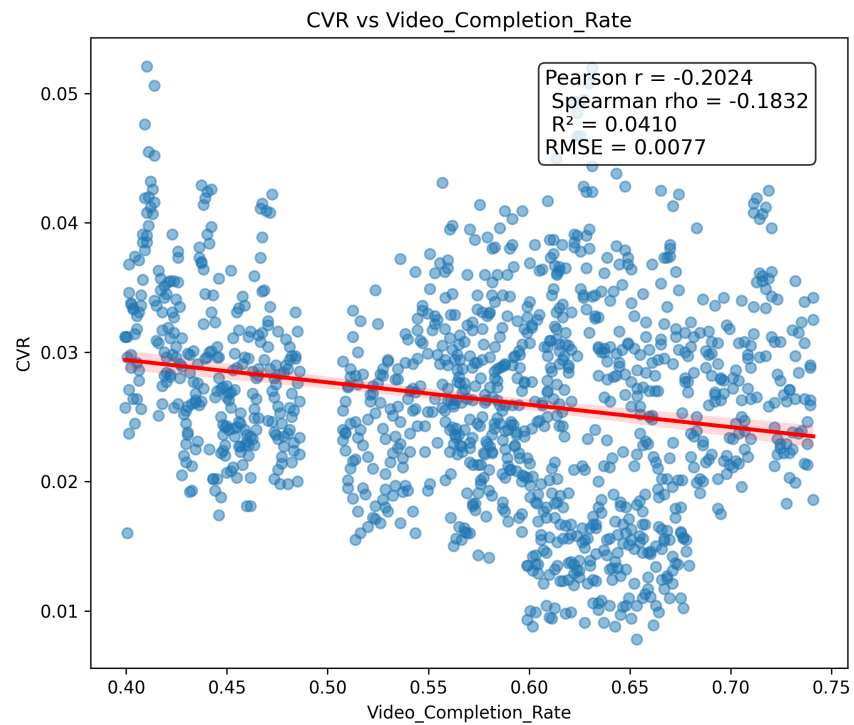


Figure 2.6: CVR vs Video Completion Rate

There is no statistically significant correlation between the conversion rate(CVR) and video completion rate.

2.5 Audience-Product Performance Analysis

Performance of existing campaigns across different audience-product pairs was also investigated.

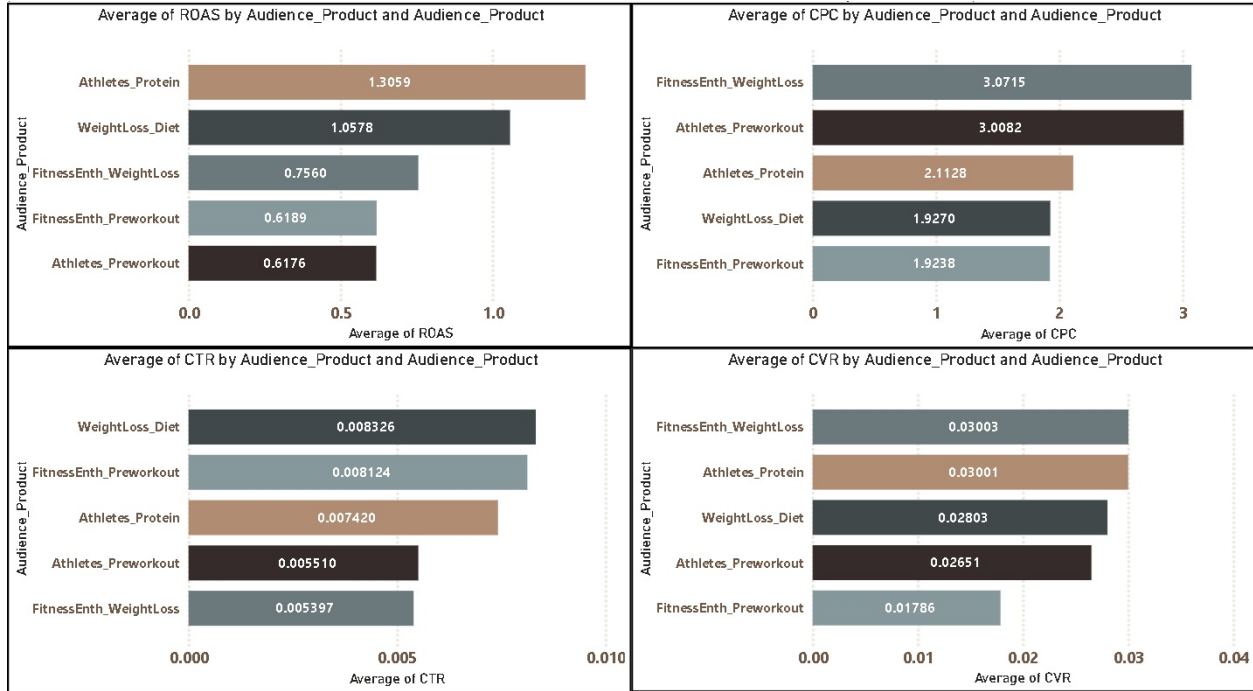


Figure 2.7: Performance Metrics by Audience-Product

- The **Athletes-Protein** combination demonstrated the strongest overall performance, achieving the highest ROAS while maintaining one of the highest conversion rates. This audience-product pairing appears to generate the greatest financial return from advertising investment.
- The **WeightLoss-Diet** combination also performed strongly, producing the second-highest ROAS, the highest click-through rate, and relatively low acquisition costs. These findings suggest that Diet campaigns are particularly effective when targeted toward individuals interested in weight loss.
- Both **Athletes-Preworkout** and **FitnessEnth-Preworkout** consistently exhibited poor ROAS despite differing levels of user engagement. This suggests that the weak financial performance of Pre-workout campaigns is not driven solely by audience selection, but may instead reflect broader issues related to the product offering, campaign messaging, pricing strategy, or post-click conversion experience.
- The **FitnessEnth-WeightLoss** combination achieved one of the highest conversion rates but also incurred the highest advertising cost per click. Although this audience converts effectively, its relatively high acquisition cost reduces overall profitability.
- Because statistically significant differences were observed across all four performance metrics, optimizing campaign performance should consider the **audience-product pairing** rather than treating audience and product category as independent marketing decisions.

From a business perspective, these findings suggest that future optimization efforts should evaluate the effectiveness of specific audience-product combinations rather than optimizing audience targeting or product campaigns independently. For example, investigating why the **WeightLoss-Diet** and **Athletes-Protein** pairings consistently outperform other combinations may reveal successful

strategies that can be applied to other campaigns. Likewise, the consistently poor performance of both **Pre-workout** audience pairings suggests that improving audience selection alone is unlikely to substantially increase profitability. Instead, optimization should focus on the entire campaign strategy surrounding these pairings, including product positioning, promotional messaging, pricing, and landing page experience.

2.6 Effect of Competitive Events

The exploratory assessment examined whether campaign performance differed during periods with and without competitive events.

A complete summary of the statistical analyses, including assumption checks, Kruskal–Wallis tests, and pairwise comparisons, is provided in Appendix B.6.

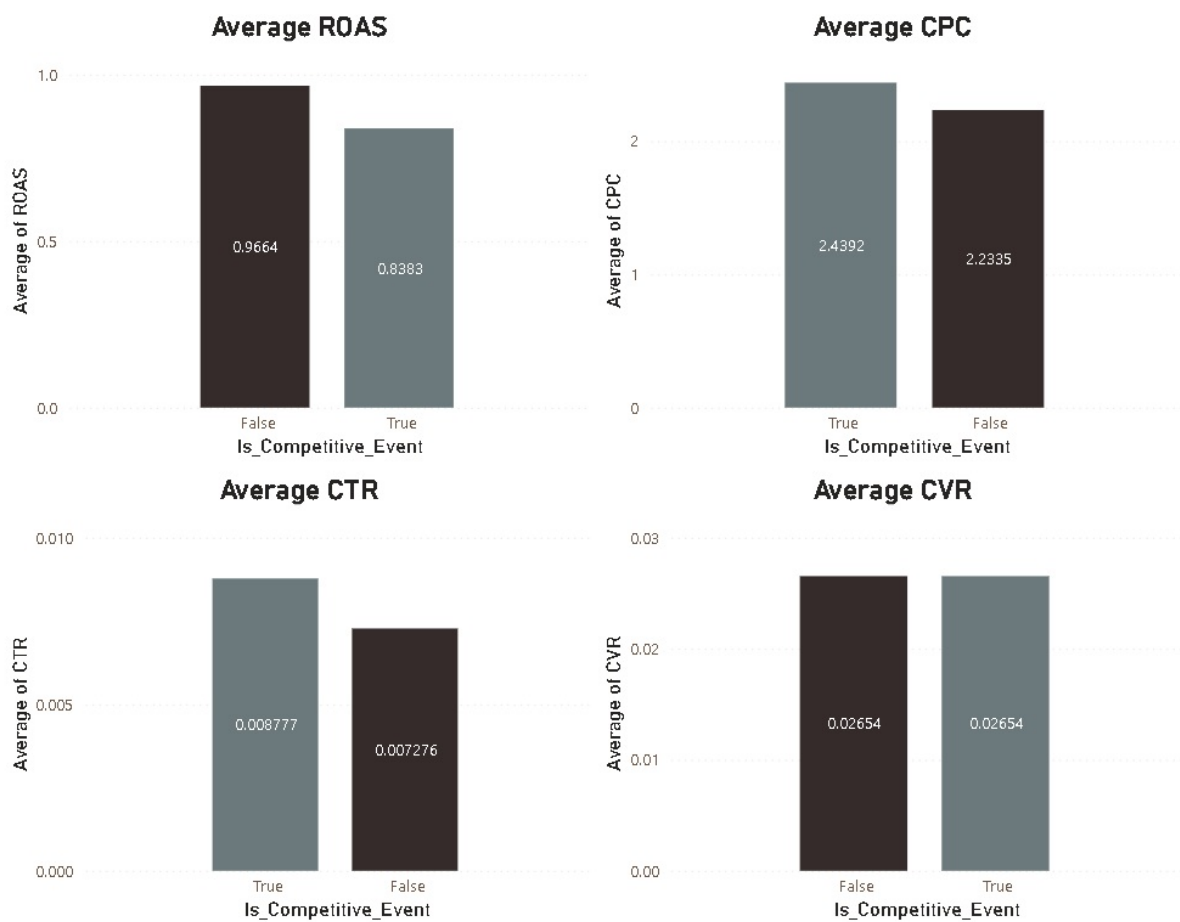


Figure 2.8: Comparison of campaign performance during competitive and non-competitive periods.

Figure 2.8 summarizes the influence of competitive events on the four primary advertising performance metrics.

Campaigns conducted during competitive events achieved a significantly lower average ROAS (0.838) than campaigns conducted outside competitive periods (0.966), indicating reduced advertising profitability (Appendix B.6.1).

Interestingly, campaigns during competitive events generated significantly higher CTR (0.88%

versus 0.73%), suggesting that advertisements attracted greater user engagement despite increased market competition (Appendix B.6.2).

However, this increase in engagement did not translate into improved purchasing behavior. Conversion rates remained statistically unchanged between competitive and non-competitive periods (approximately 2.65%), indicating that users who clicked advertisements were equally likely to complete a purchase regardless of competitive conditions (Appendix B.6.3).

Competitive events were also associated with significantly higher CPC (\$2.44 versus \$2.23), indicating that advertisers paid more to acquire each click (Appendix B.6.4).

Taken together, these findings suggest that competitive events primarily reduce campaign profitability by increasing advertising costs rather than by reducing customer purchase intent. Although advertisements attracted more clicks during competitive periods, the higher acquisition cost combined with unchanged conversion performance resulted in significantly lower ROAS.

From a business perspective, campaigns scheduled during competitive periods may require stronger creative assets, more precise audience targeting, or more compelling promotional offers to justify the higher advertising costs. Alternatively, campaigns focused primarily on maximizing profitability may benefit from allocating a larger portion of the advertising budget to less competitive periods where comparable conversion rates can be achieved at lower acquisition costs.

Appendix A

Data Quality and Initial Exploratory Assessment

A.1 Data Quality Assessment

Prior to conducting business intelligence analysis, the advertising campaign dataset was evaluated to ensure its quality, reliability, and suitability for analysis. The objective of this assessment is to identify potential data quality issues, validate the performance metrics used throughout the project, examine unusual observations, and establish a set of baseline metrics that will guide subsequent business analysis.

The dataset contains daily advertising campaign performance across multiple platforms, regions, target audiences, product categories, and creative types during the first quarter of 2024.

A.1.1 Data Type

The data types of all variables were first examined to verify that each column was stored using an appropriate data type. As shown in *Fig. A.1*, numerical variables were stored as either `int64` or `float64`, categorical variables as `object`, the `Date` column as `datetime64[ns]`, and the `Is_Competitive_Event` variable as a boolean.

```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3240 entries, 0 to 3239
Data columns (total 17 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Date                  3240 non-null  datetime64[ns]
1   Platform              3240 non-null  object
2   Campaign              3240 non-null  object
3   Region                3240 non-null  object
4   Spend                 3240 non-null  float64
5   CPM                   3240 non-null  float64
6   Impressions           3240 non-null  int64
7   Frequency             3240 non-null  float64
8   Clicks                3240 non-null  int64
9   Purchases             3240 non-null  int64
10  Revenue               3228 non-null  float64
11  Product_Category      3240 non-null  object
12  Target_Audience       3240 non-null  object
13  Creative_Type          3240 non-null  object
14  Video_Completion_Rate  1228 non-null  float64
15  Customer_LTV           3240 non-null  float64
16  Is_Competitive_Event   3240 non-null  bool
```

Figure A.1: Checking datatypes of the dataframe

No inconsistencies in data types were identified; therefore, no type conversions were required.

A.1.2 Duplicate Records

The dataset was examined for duplicate observations.

```

We checked for duplicated rows and found nothing.

> print("Number of duplicated rows: ", df.duplicated().sum(), "rows")
[229] ✓ 0.0:
... Number of duplicated rows: 0 rows

```

Figure A.2: Checking for duplicate records

No duplicate records were identified as shown in *Fig. A.2*.

A.1.3 Data Completeness

The dataset was examined for missing values to determine whether incomplete observations could affect subsequent analyses.

- **Revenue**

A sample of these datapoints are shown in *Fig. A.3*. It can be seen that twenty observations (approximately 0.6% of the dataset) contained missing revenue values.

2. Revenue

There seems to be some missing values for the 'Revenue' column as well. This column is very important as it is used as the basis for the KPI metrics like ROAS, CPA, etc.

```
df[df['Revenue'].isna()].head()
```

	Date	Platform	Campaign	Region	Spend	CPM	Impressions	Frequency	Clicks	Purchases	Revenue	Product_Category	Target_Audience	Creative_Type	Video_Completion_Rate	Customer_LTV	Is_Competitor
194	2024-01-06	Google	Google_Athletes_Search_Protein_001	Northeast	1225.52	18.09	67745	2.27	1236	35	NaN	Protein	Athletes	Search	NaN	940.8	
589	2024-01-17	Google	Google_Athletes_Search_Protein_001	South	978.48	15.24	64204	2.03	979	32	NaN	Protein	Athletes	Search	NaN	940.8	
1413	2024-02-09	FB	FB_FitnessEnth_Carousel_WeightLoss_001	South	1909.27	12.47	153133	3.61	823	24	NaN	WeightLoss	FitnessEnth	Carousel	NaN	448.5	
1658	2024-02-16	FB	FB_Athletes_Video_Protein_001	Northeast	974.65	12.55	77649	2.08	505	16	NaN	Protein	Athletes	Video	0.4413	772.8	
1661	2024-02-16	FB	FB_Athletes_Image_Previewout_002	South	913.94	12.55	72812	4.25	243	6	NaN	Previewout	Athletes	Image	NaN	538.2	
1682	2024-02-16	TT	TT_Athletes_Video_Protein_001	Northeast	920.33	10.46	87985	2.92	513	20	NaN	Protein	Athletes	Video	0.6728	604.8	
1738	2024-02-18	FB	FB_FitnessEnth_Carousel_WeightLoss_001	Northeast	2696.87	15.09	178704	2.17	881	28	NaN	WeightLoss	FitnessEnth	Carousel	NaN	448.5	

Figure A.3: Sample datapoints with missing revenues.

Because these represented a very small proportion of the data and no systematic pattern of missingness was identified, the corresponding observations were removed as shown in *Fig. A.4* prior to analysis.

- **Video Completion Rate**

A large number of missing values were initially observed in the *Video Completion Rate* variable. Around 62% of the total dataset!

Further investigation revealed that this metric is only applicable to **campaigns using video advertisements**. After restricting the dataframe to video creatives, **only 212 missing observations remained** as shown in *Fig. A.6*.

```

Because the proportion of missing values is very small, removing these observations is unlikely to introduce significant bias or substantially reduce the available information. Consequently, these rows will be removed from the dataset for analyses involving the Revenue variable.

print("BEFORE DROPPING: Number of rows with missing revenue:", len(df[df['Revenue'].isna()]))
df.dropna(subset=["Revenue"], inplace=True)
print("AFTER DROPPING: Number of rows with missing revenue:", len(df[df['Revenue'].isna()]))
225 ✓ 0.0s Python
BEFORE DROPPING: Number of rows with missing revenue: 20
AFTER DROPPING: Number of rows with missing revenue: 0

```

Figure A.4: Dropped datapoints with missing revenues.

```

df.info()
226 ✓ 0.0s
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3240 entries, 0 to 3239
Data columns (total 17 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Date                  3240 non-null   datetime64[ns]
1   Platform              3240 non-null   object
2   Campaign              3240 non-null   object
3   Region               3240 non-null   object
4   Spend                3240 non-null   float64
5   CPW                  3240 non-null   float64
6   Impressions          3240 non-null   int64
7   Frequency            3240 non-null   float64
8   Clicks               3240 non-null   int64
9   Purchases            3240 non-null   int64
10  Revenue              3220 non-null   float64
11  Product_Category     3240 non-null   object
12  Target_Audience     3240 non-null   object
13  Creative_Type        3240 non-null   object
14  Video_Completion_Rate 1228 non-null   float64
15  Customer_City        3240 non-null   float64
16  Is_Competitive_Event  3240 non-null   bool

```

Figure A.5: video_completion_rate missing values

Consequently, these missing values were treated as **structurally missing** rather than data quality issues, and analyses involving this variable will use only the relevant subset of campaigns.

A.1.4 Logical Consistency Checks

Several validation checks were performed to identify logically inconsistent observations.

0 Clicks with non-zero purchase

I noticed that there were rows with 0 clicks but has non-zero Purchases, which I thought was weird at first. How are they able to purchase without clicking the ad.

Apparently, this happens in real world scenarios and they are called **view-through conversions**. People see the ad, they don't click it then later that evening, they remembered the brand and typed the website into google and bought the product.

But, at this point, I was wondering how they know which ad actually influenced the purchased without click or what if the buyer interacted with multiple ads before purchasing (cross-channel conversions). Apparently, this is related to **attribution**, and there are multiple attribution models like **last_click attribution**, **first-click attribution**, **linear attribution**, etc.. which is not part of this data cleaning process.

So, for now, I'll just assume that this dataset was collected using a certain attribution model and leave it at that. This justifies the 0 clicks with non-zero purchase.

```

df_video_type = df[df['Creative_Type'] == 'Video']

df_video_type.info()
<class 'pandas.core.frame.DataFrame'>
Index: 1440 entries, 0 to 3239
Data columns (total 17 columns):
#   Column              Non-Null Count  Dtype
---  -
0   Date                 1440 non-null   datetime64[ns]
1   Platform             1440 non-null   object
2   Campaign             1440 non-null   object
3   Region              1440 non-null   object
4   Spend               1440 non-null   float64
5   CPM                 1440 non-null   float64
6   Impressions         1440 non-null   int64
7   Frequency            1440 non-null   float64
8   Clicks              1440 non-null   int64
9   Purchases           1440 non-null   int64
10  Revenue             1424 non-null   float64
11  Product_Category    1440 non-null   object
12  Target_Audience    1440 non-null   object
13  Creative_Type       1440 non-null   object
14  Video_Completion_Rate 1228 non-null   float64
15  Customer_LTV        2440 non-null   float64
16  Is_Competitive_Event 1440 non-null   bool

```

Figure A.6: DataFrame filtered to video creative types

df[(df['clicks'] == 0)]

	Date	Platform	Campaign	Region	Spend	CPM	Impressions	Frequency	Clicks	Purchases	Revenue	Product_Category	Target_Audience	Creative_Type	Video_Completion_Rate	Customer_LTV	Is_Competitive_Event
51	2024-01-02	Google	Google_Athlete_Search_Protein_001	Midwest	1045.31	15.01	0	3.09	0	0	2021.74	Protein	Athletes	Search	NaN	940.8	
86	2024-01-03	Google	Google_Athlete_Search_Protein_001	Northeast	1303.24	15.03	86709	2.61	0	41	3593.14	Protein	Athletes	Search	NaN	940.8	
92	2024-01-03	Google	Google_Fitness_Display_Previewout_001	West	469.88	15.03	31262	2.57	0	13	715.20	Previewout	FitnessEnth	Display	NaN	546.0	
94	2024-01-03	Google	Google_Fitness_Display_Previewout_001	Northeast	921.64	15.03	61320	3.77	0	12	691.31	Previewout	FitnessEnth	Display	NaN	546.0	
122	2024-01-04	Google	Google_Athlete_Search_Protein_001	Northeast	642.88	15.04	42730	3.42	0	16	1388.57	Protein	Athletes	Search	NaN	940.8	
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
3047	2024-03-25	Google	Google_Fitness_Display_Previewout_001	Midwest	988.78	16.26	60810	3.28	0	5	307.60	Previewout	FitnessEnth	Display	NaN	546.0	
3075	2024-03-26	Google	Google_Athlete_Search_Protein_001	Midwest	950.82	18.72	50802	2.51	0	7	562.38	Protein	Athletes	Search	NaN	940.8	
3109	2024-03-26	Google	Google_Athlete_Search_Protein_001	South	1181.40	18.73	63063	2.41	0	8	652.57	Protein	Athletes	Search	NaN	940.8	

Figure A.7: Sample datapoints with 0 clicks but non-zero purchase

Side Note: After the data cleaning, all rows with zero clicks belong to the Google platform. (For the time being, this is just an observation. I'm noting it here so I don't forget to investigate it later.)

0 Impressions with non-zero Cost-Per-Mile(CPM)

Cost Per Mille (CPM) measures the advertising cost required to generate 1,000 impressions and is commonly used to evaluate the cost efficiency of awareness campaigns.

$$\text{CPM} = \frac{\text{Spend}}{\text{Impressions}} \times 1000 \quad (\text{A.1})$$

Because impressions appear in the denominator, CPM is undefined when the number of impressions is zero.

During the data quality assessment, 47 observations (approximately 1.45% of the dataset) were identified with **Impressions** = 0 but non-zero with CPM values.

Furthermore, all of these observations originated from the Midwest region as shown in *Fig. A.8*, suggesting a systematic reporting inconsistency rather than isolated data entry errors.

Since these observations violate the definition of CPM and the 0 impressions value could affect the calculation of other performance metrics, such as CTR and CVR, they were removed from the dataset prior to analysis.

	Date	Platform	Campaign	Region	Spend	CPM	Impressions	Frequency	Clicks	Purchases	Revenue	Product Category	Target Audience	Creative Type	Video Completion Rate	Customer LTV	Is Competitive
51	2024-01-02	Google	Google_Athlete_Search_Protein_001	Midwest	845.31	15.01	0	3.89	0	0	2021.74	Protein	Athletes	Search	NaN	940.80	
199	2024-01-05	Google	Google_WeightLoss_Search_Diet_001	Midwest	867.90	18.08	0	4.16	0	0	1542.29	Diet	WeightLoss	Search	NaN	393.12	
251	2024-01-07	TT	TT_WeightLoss_Video_Diet_001	Midwest	619.73	12.97	0	2.11	0	0	3362.89	Diet	WeightLoss	Video	0.6101	252.72	
259	2024-01-08	FB	FB_Athlete_Image_Previewout_002	Midwest	745.96	12.08	0	2.83	0	0	527.76	Previewout	Athletes	Image	NaN	538.20	
311	2024-01-09	Google	Google_FitnessEnth_Display_Previewout_001	Midwest	616.50	15.12	0	2.42	0	0	518.51	Previewout	FitnessEnth	Display	NaN	546.00	

Figure A.8: Sample rows that has 0 impressions with non-zero CPM

```

mask = (df["Impressions"] == 0) & (df["CPM"] > 0)

print("Rows before removal:", mask.sum())

df.drop(df[mask].index, inplace=True)

mask = (df["Impressions"] == 0) & (df["CPM"] > 0)
print("Rows after removal:", mask.sum())

```

[396] ✓ 0.0s

... Rows before removal: 47
Rows after removal: 0

Figure A.9: Dropped Rows that has 0 impressions with non-zero CPM

A.1.5 Outlier Assessment

Boxplots were used to identify potential outliers across the numerical variables as shown in *Fig. A.10*. Severe outliers were identified for the Spend column and will be investigated in this section.

Spend severe outliers investigation

First, I listed the 5 datapoints with severe **Spend** outliers as shown in *Fig. A.12*. I noticed that in this 5 datapoints, the **CPM do not match the Ad Spend and Impression**. This tells me that there is a possibility that there are more rows with mismatched CPM.

So, I checked and found out that only these 5 rows have mismatched CPM as shown in *Fig. A.11*, after dropping the rows from earlier (the ones with 0 impression, clicks and purchases).

Then, I plotted the **Spend** boxplots of these Campaign Ads.

From this analysis, I noticed the following consistencies:

1. The reported CPM values (from datapoints with severe **Spend** outliers) were exactly one-hundredth of the values calculated from the corresponding Spend and Impressions.
2. The calculated CPM values do not match the 'Impressions' and 'Spend' columns, and this is limited only to the 5 reported rows with severe 'Spend' outliers.
3. Inspection of the affected campaigns showed that the unusually large Spend values occurred only on a single day and were not associated with competitive events.

Since the discrepancy was systematic (a factor of 100) and limited to these five observations, the Spend values were divided by 100 to restore consistency with the reported CPM values.

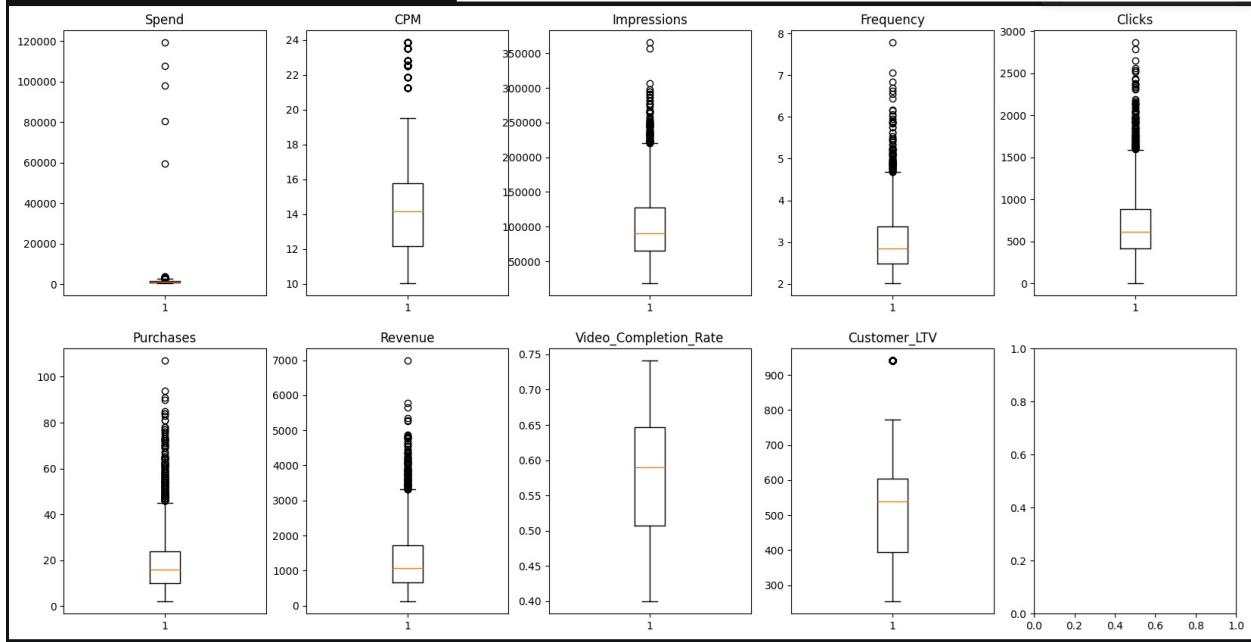


Figure A.10: Boxplots for all numerical columns of the dataframe

A.1.6 Data Cleaning Summary

The data cleaning process consisted of:

- Verified variable data types
- Verifying duplicate observations
- Removing observations(20 datapoints) with missing revenue.
- Retaining structurally missing Video Completion Rate values.
- Retaining valid view-through conversions.
- Removing observations(47 datapoints) with 0 clicks,impressions and purchases with non-zero purchase.
- Addressed severe **Spend** outliers.

Overall, the dataset was determined to be sufficiently complete and reliable for subsequent analysis.



Figure A.11: Number of rows with CPM mismatch

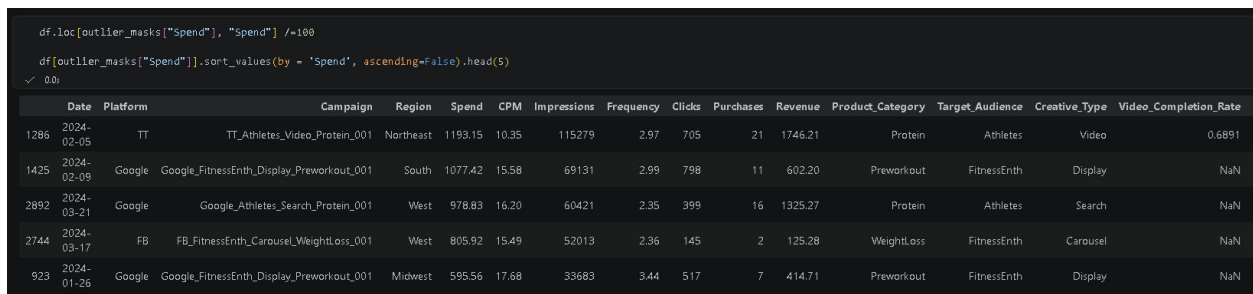


Figure A.12: Datapoints with severe Spend outliers

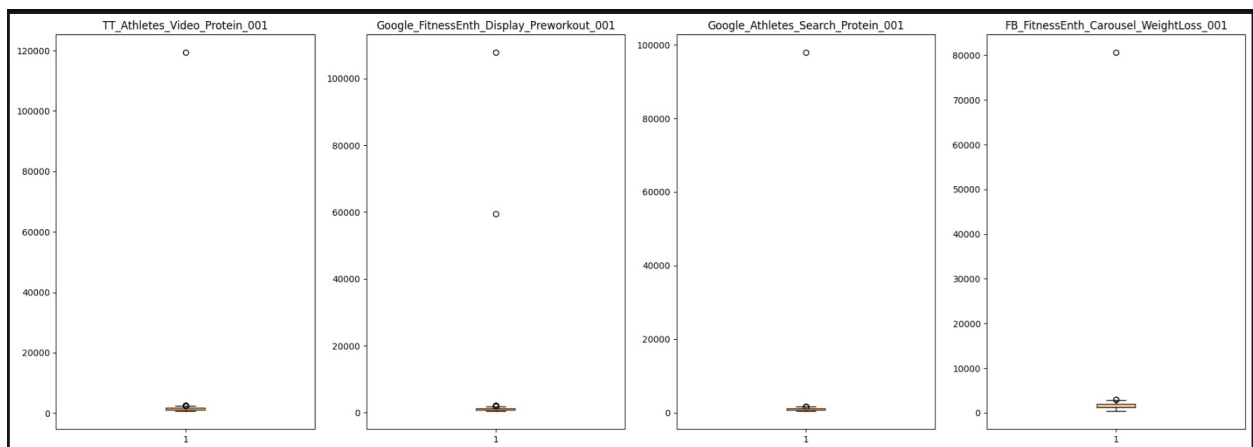


Figure A.13: Spend boxplots for campaign ads with severe Spend outliers

Appendix B

Statistical Test Results

B.1 Channel performance analysis

B.1.1 Return on Advertising Spend (ROAS)

Descriptive Statistics

Table B.1: Descriptive statistics for ROAS by platform.

Platform	Mean	Median	Std. Dev.	Count
Facebook	0.8162	0.7499	0.4225	1054
Google	1.1057	0.9456	0.6495	1053
TikTok	0.9602	0.8625	0.4950	1066

Assumption Tests

Shapiro–Wilk Test

Platform	p-value
Facebook	< 0.0001
Google	< 0.0001
TikTok	< 0.0001

All platform distributions significantly deviated from normality ($p < 0.05$).

Levene’s Test

Statistic	Value
p-value	< 0.0001

The variability of **ROAS** differed **significantly** across platforms.

Kruskal–Wallis Test

Statistic	Value
H-statistic	106.6989
p-value	< 0.0001

The Kruskal–Wallis test indicates statistically significant differences in ROAS across advertising platforms.

Dunn Post-hoc Test

Table B.2: Pairwise Dunn test p-values (Bonferroni adjusted).

	Facebook	Google	TikTok
Facebook	–	< 0.0001	< 0.0001
Google	< 0.0001	–	0.0002
TikTok	< 0.0001	0.0002	–

Summary

- The highest average ROAS was achieved by Google (1.1057).
- The lowest average ROAS was observed for FB (0.8162).
- Differences of ROAS between platforms are statistically significant ($p < 0.05$).

B.1.2 Click-Through Rate (CTR)**Descriptive Statistics**

Table B.3: Descriptive statistics for CTR by platform.

Platform	Mean	Median	Std. Dev.	Count
Facebook	0.0057	0.0056	0.0021	1054
Google	0.0100	0.0098	0.0052	1053
TikTok	0.0063	0.0062	0.0018	1066

Assumption Tests**Shapiro–Wilk Test**

Platform	p-value
Facebook	< 0.0001
Google	< 0.0001
TikTok	< 0.0001

All platform distributions significantly deviated from normality ($p < 0.05$).

Levene's Test

Statistic	Value
p-value	< 0.0001

The variability of CTR **differed significantly across platforms**.

Kruskal–Wallis Test

Statistic	Value
H-statistic	634.6820
p-value	< 0.0001

The Kruskal–Wallis test indicates statistically significant differences in CTR across advertising platforms.

Dunn Post-hoc Test

Table B.4: Pairwise Dunn test p-values (Bonferroni adjusted).

	Facebook	Google	TikTok
Facebook	–	< 0.0001	< 0.0001
Google	< 0.0001	–	< 0.0001
TikTok	< 0.0001	< 0.0001	–

Summary

- The highest average CTR was achieved by Google (0.0100).
- The lowest average CTR was observed for FB (0.0057).
- Differences of CTR between platforms are statistically significant ($p < 0.05$).

B.1.3 Conversion Rate (CVR)

Descriptive Statistics

Table B.5: Descriptive statistics for CVR by platform.

Platform	Mean	Median	Std. Dev.	Count
Facebook	0.0287	0.0277	0.0076	1054
Google	0.0254	0.0254	0.0082	954
TikTok	0.0254	0.0258	0.0081	1066

Assumption Tests

Shapiro–Wilk Test

Platform	p-value
Facebook	< 0.0001
Google	< 0.0001
TikTok	< 0.0001

All platform distributions **significantly deviated from normality** ($p < 0.05$).

Levene’s Test

Statistic	Value
p-value	0.0072

The variability of CVR **differed significantly across platforms**.

Kruskal–Wallis Test

Statistic	Value
H-statistic	96.7108
p-value	< 0.0001

The Kruskal–Wallis test indicates **that at least one platform differs significantly** in CVR.

Dunn Post-hoc Test

Table B.6: Pairwise Dunn test p-values (Bonferroni adjusted).

	Facebook	Google	TikTok
Facebook	–	< 0.0001	< 0.0001
Google	< 0.0001	–	1.0000
TikTok	< 0.0001	1.0000	–

Facebook differed significantly from both Google and TikTok, whereas **no statistically significant difference** was observed between Google and TikTok.

Summary

- The **highest average CVR** was achieved by **Facebook** (0.0287).
- The **lowest average CVR** was observed for **Google** (0.0254), although Google and TikTok had nearly identical average CVRs.
- **Facebook differed significantly** from both Google and TikTok, while **no statistically significant difference** was observed between Google and TikTok.

B.1.4 Cost Per Click (CPC)

Descriptive Statistics

Table B.7: Descriptive statistics for CPC by platform.

Platform	Mean	Median	Std. Dev.	Count
Facebook	2.8374	2.5049	1.3116	1054
Google	1.8747	1.5936	0.9076	954
TikTok	1.9840	1.8452	0.6718	1066

Assumption Tests

Shapiro–Wilk Test

Platform	p-value
Facebook	< 0.0001
Google	< 0.0001
TikTok	< 0.0001

All platform distributions significantly deviated from normality ($p < 0.05$).

Levene's Test

Statistic	Value
p-value	< 0.0001

The variability of **CPC differed significantly** across platforms.

Kruskal–Wallis Test

Statistic	Value
H-statistic	447.1995
p-value	< 0.0001

The Kruskal–Wallis test indicates statistically significant differences in CPC across advertising platforms.

Dunn Post-hoc Test

Table B.8: Pairwise Dunn test p-values (Bonferroni adjusted).

	Facebook	Google	TikTok
Facebook	–	< 0.0001	< 0.0001
Google	< 0.0001	–	< 0.0001
TikTok	< 0.0001	< 0.0001	–

Summary

- The highest average CPC was achieved by Facebook (2.8374).
- The lowest average CPC was observed for Google (1.8747).
- Differences of CPC between platforms are statistically significant ($p < 0.05$).

B.2 Regional performance comparison

B.2.1 Return on Advertising Spend (ROAS)

Descriptive Statistics

Table B.9: Descriptive statistics for ROAS by region.

Region	Mean	Median	Std. Dev.	Count
Midwest	0.8337	0.6951	0.4940	760
Northeast	1.0330	0.8678	0.5960	803
South	0.9605	0.8535	0.4868	803
West	1.0083	0.8915	0.5662	807

Assumption Tests

Shapiro–Wilk Test

Region	p-value
Midwest	< 0.0001
Northeast	< 0.0001
South	< 0.0001
West	< 0.0001

All regional distributions significantly deviated from normality ($p < 0.05$).

Levene’s Test

Statistic	Value
p-value	< 0.0001

The variability of ROAS differed significantly across regions.

Kruskal–Wallis Test

Statistic	Value
H-statistic	69.2321
p-value	< 0.0001

The Kruskal–Wallis test indicates that at least one region differs significantly in ROAS.

Dunn Post-hoc Test

Table B.10: Pairwise Dunn test p-values (Bonferroni adjusted).

	Midwest	Northeast	South	West
Midwest	–	< 0.0001	< 0.0001	< 0.0001
Northeast	< 0.0001	–	1.0000	1.0000
South	< 0.0001	1.0000	–	1.0000
West	< 0.0001	1.0000	1.0000	–

The Midwest **differed significantly** from the Northeast, South, and West, whereas **no statistically significant differences** were observed among the Northeast, South, and West.

Summary

- The highest average ROAS was achieved by the Northeast (1.0330).
- The lowest average ROAS was observed for the Midwest (0.8337).
- The Midwest differed significantly from all other regions, while no statistically significant differences were observed among the Northeast, South, and West.

B.2.2 Click-Through Rate (CTR)

Descriptive Statistics

Table B.11: Descriptive statistics for CTR by region.

Region	Mean	Median	Std. Dev.	Count
Midwest	0.0074	0.0066	0.0039	760
Northeast	0.0072	0.0064	0.0039	803
South	0.0074	0.0067	0.0038	803
West	0.0073	0.0067	0.0038	807

Assumption Tests

Shapiro–Wilk Test

Region	p-value
Midwest	< 0.0001
Northeast	< 0.0001
South	< 0.0001
West	< 0.0001

All regional distributions **significantly deviated** from normality ($p < 0.05$).

Levene's Test

Statistic	Value
p-value	0.8844

The variability of CTR **did not differ significantly** across regions.

Kruskal–Wallis Test

Statistic	Value
H-statistic	0.9848
p-value	0.8049

The Kruskal–Wallis test indicates **no statistically significant differences** in CTR across regions.

Summary

- The highest average CTR was achieved by the Midwest (0.0074), although the regional averages were very similar.
- The lowest average CTR was observed for the Northeast (0.0072).
- No statistically significant differences in CTR were found between regions ($p \geq 0.05$).

B.2.3 Conversion Rate (CVR)**Descriptive Statistics**

Table B.12: Descriptive statistics for CVR by region.

Region	Mean	Median	Std. Dev.	Count
Midwest	0.0229	0.0236	0.0073	737
Northeast	0.0284	0.0288	0.0086	771
South	0.0270	0.0270	0.0071	785
West	0.0277	0.0267	0.0081	781

Assumption Tests**Shapiro–Wilk Test**

Region	p-value
Midwest	< 0.0001
Northeast	< 0.0001
South	< 0.0001
West	< 0.0001

All regional distributions **significantly deviated** from normality ($p < 0.05$).

Levene’s Test

Statistic	Value
p-value	< 0.0001

The variability of CVR **differed significantly** across regions.

Kruskal–Wallis Test

Statistic	Value
H-statistic	190.4208
p-value	< 0.0001

The Kruskal–Wallis test indicates that at least one region **differs significantly** in CVR.

Dunn Post-hoc Test

Table B.13: Pairwise Dunn test p-values (Bonferroni adjusted).

	Midwest	Northeast	South	West
Midwest	–	< 0.0001	< 0.0001	< 0.0001
Northeast	< 0.0001	–	0.0057	0.0396
South	< 0.0001	0.0057	–	1.0000
West	< 0.0001	0.0396	1.0000	–

The Midwest **differed significantly** from all other regions. The Northeast also differed significantly from both the South and the West, whereas **no statistically significant difference** was observed between the South and the West.

Summary

- The highest average CVR was achieved by the Northeast (0.0284).
- The lowest average CVR was observed for the Midwest (0.0229).
- The Midwest differed significantly from all other regions, while the South and West did not differ significantly from one another.

B.2.4 Cost Per Click (CPC)

Descriptive Statistics

Table B.14: Descriptive statistics for CPC by region.

Region	Mean	Median	Std. Dev.	Count
Midwest	2.2477	1.9809	1.1166	737
Northeast	2.2475	1.9919	1.0984	771
South	2.2374	2.0091	1.0791	785
West	2.2386	1.9872	1.0686	781

Assumption Tests

Shapiro–Wilk Test

Region	p-value
Midwest	< 0.0001
Northeast	< 0.0001
South	< 0.0001
West	< 0.0001

All regional distributions **significantly deviated** from normality ($p < 0.05$).

Levene’s Test

Statistic	Value
p-value	0.9460

The variability of CPC did **not differ significantly** across regions.

Kruskal–Wallis Test

Statistic	Value
H-statistic	0.0346
p-value	0.9983

The Kruskal–Wallis test indicates **no statistically significant differences** in CPC across regions.

Summary

- The highest average CPC was observed in the Midwest (2.2477), although the regional averages were nearly identical.
- The lowest average CPC was observed in the South (2.2374).
- No statistically significant differences in CPC were found between regions ($p \geq 0.05$).

B.3 Creative performance analysis

B.3.1 Return on Advertising Spend (ROAS)

Descriptive Statistics

Table B.15: Descriptive statistics for ROAS by creative type.

Creative Type	Mean	Median	Std. Dev.	Count
Carousel	0.7560	0.6572	0.4480	355
Display	0.6626	0.6194	0.2685	348
Image	0.6176	0.5656	0.2749	349
Search	1.3244	1.2455	0.6708	705
Video	0.9886	0.9058	0.4730	1416

Assumption Tests

Shapiro–Wilk Test

Creative Type	p-value
Carousel	< 0.0001
Display	< 0.0001
Image	< 0.0001
Search	< 0.0001
Video	< 0.0001

All creative type distributions significantly deviated from normality ($p < 0.05$).

Levene's Test

Statistic	Value
p-value	< 0.0001

The variability of **ROAS** differed significantly across creative types.

Kruskal–Wallis Test

Statistic	Value
H-statistic	606.8404
p-value	< 0.0001

The Kruskal–Wallis test indicates statistically significant differences in ROAS across creative types.

Dunn Post-hoc Test

Table B.16: Pairwise Dunn test p-values (Bonferroni adjusted).

	Carousel	Display	Image	Search	Video
Carousel	–	0.2527	0.0008	< 0.0001	< 0.0001
Display	0.2527	–	0.8940	< 0.0001	< 0.0001
Image	0.0008	0.8940	–	< 0.0001	< 0.0001
Search	< 0.0001	< 0.0001	< 0.0001	–	< 0.0001
Video	< 0.0001	< 0.0001	< 0.0001	< 0.0001	–

Summary

- The highest average ROAS was achieved by **Search** (1.3244).
- The lowest average ROAS was observed for **Image** (0.6176).
- **Search** differed significantly from all other creative types. No statistically significant differences were observed between **Carousel** and **Display**, or between **Display** and **Image**.

B.3.2 Click-Through Rate (CTR)

Descriptive Statistics

Table B.17: Descriptive statistics for CTR by creative type.

Creative Type	Mean	Median	Std. Dev.	Count
Carousel	0.0054	0.0051	0.0021	355
Display	0.0100	0.0098	0.0054	348
Image	0.0055	0.0052	0.0022	349
Search	0.0100	0.0099	0.0050	705
Video	0.0063	0.0061	0.0018	1416

Assumption Tests

Shapiro–Wilk Test

Creative Type	p-value
Carousel	< 0.0001
Display	< 0.0001
Image	< 0.0001
Search	< 0.0001
Video	< 0.0001

All creative type distributions significantly deviated from normality ($p < 0.05$).

Levene’s Test

Statistic	Value
p-value	< 0.0001

The variability of **CTR differed significantly** across creative types.

Kruskal–Wallis Test

Statistic	Value
H-statistic	658.6765
p-value	< 0.0001

The Kruskal–Wallis test indicates statistically significant differences in CTR across creative types.

Dunn Post-hoc Test

Table B.18: Pairwise Dunn test p-values (Bonferroni adjusted).

	Carousel	Display	Image	Search	Video
Carousel	–	< 0.0001	1.0000	< 0.0001	< 0.0001
Display	< 0.0001	–	< 0.0001	1.0000	< 0.0001
Image	1.0000	< 0.0001	–	< 0.0001	< 0.0001
Search	< 0.0001	1.0000	< 0.0001	–	< 0.0001
Video	< 0.0001	< 0.0001	< 0.0001	< 0.0001	–

Summary

- The highest average CTR was achieved by **Display** (0.0100), closely followed by **Search** (0.0100).
- The lowest average CTR was observed for **Carousel** (0.0054).
- Display and Search did not differ significantly, while Carousel and Image also showed no statistically significant difference. All remaining pairwise comparisons were statistically significant.

B.3.3 Conversion Rate (CVR)

Descriptive Statistics

Table B.19: Descriptive statistics for CVR by creative type.

Creative Type	Mean	Median	Std. Dev.	Count
Carousel	0.0300	0.0289	0.0078	355
Display	0.0178	0.0169	0.0054	310
Image	0.0265	0.0252	0.0081	349
Search	0.0291	0.0284	0.0066	644
Video	0.0264	0.0266	0.0079	1416

Assumption Tests

Shapiro–Wilk Test

Creative Type	p-value
Carousel	< 0.0001
Display	< 0.0001
Image	< 0.0001
Search	< 0.0001
Video	0.0001

All creative type distributions significantly deviated from normality ($p < 0.05$).

Levene's Test

Statistic	Value
p-value	< 0.0001

The variability of **CVR** differed significantly across creative types.

Kruskal–Wallis Test

Statistic	Value
H-statistic	489.5054
p-value	< 0.0001

The Kruskal–Wallis test indicates statistically significant differences in CVR across creative types.

Dunn Post-hoc Test

Table B.20: Pairwise Dunn test p-values (Bonferroni adjusted).

	Carousel	Display	Image	Search	Video
Carousel	–	< 0.0001	< 0.0001	1.0000	< 0.0001
Display	< 0.0001	–	< 0.0001	< 0.0001	< 0.0001
Image	< 0.0001	< 0.0001	–	< 0.0001	1.0000
Search	1.0000	< 0.0001	< 0.0001	–	< 0.0001
Video	< 0.0001	< 0.0001	1.0000	< 0.0001	–

Carousel and Search did not differ significantly, while Image and Video also showed no statistically significant difference. All remaining pairwise comparisons were statistically significant.

Summary

- The **highest average CVR** was achieved by **Carousel** (0.0300), closely followed by **Search** (0.0291).
- The **lowest average CVR** was observed for **Display** (0.0178).
- Carousel and Search exhibited statistically similar CVRs, while Image and Video also did not differ significantly. All other pairwise comparisons showed statistically significant differences.

B.3.4 Cost Per Click (CPC)

Descriptive Statistics

Table B.21: Descriptive statistics for CPC by creative type.

Creative Type	Mean	Median	Std. Dev.	Count
Carousel	3.0715	2.6470	1.4731	355
Display	1.8540	1.5707	0.9053	310
Image	3.0082	2.6030	1.4140	349
Search	1.8847	1.6315	0.9092	644
Video	2.0942	1.9452	0.7485	1416

Assumption Tests

Shapiro–Wilk Test

Creative Type	p-value
Carousel	< 0.0001
Display	< 0.0001
Image	< 0.0001
Search	< 0.0001
Video	< 0.0001

All creative type distributions significantly deviated from normality ($p < 0.05$).

Levene’s Test

Statistic	Value
p-value	< 0.0001

The variability of **CPC differed significantly** across creative types.

Kruskal–Wallis Test

Statistic	Value
H-statistic	409.9110
p-value	< 0.0001

The Kruskal–Wallis test indicates statistically significant differences in CPC across creative types.

Dunn Post-hoc Test

Table B.22: Pairwise Dunn test p-values (Bonferroni adjusted).

	Carousel	Display	Image	Search	Video
Carousel	–	< 0.0001	1.0000	< 0.0001	< 0.0001
Display	< 0.0001	–	< 0.0001	1.0000	< 0.0001
Image	1.0000	< 0.0001	–	< 0.0001	< 0.0001
Search	< 0.0001	1.0000	< 0.0001	–	< 0.0001
Video	< 0.0001	< 0.0001	< 0.0001	< 0.0001	–

Carousel and Image did not differ significantly, while Display and Search also exhibited statistically similar CPC values. All remaining pairwise comparisons were statistically significant.

Summary

- The **highest average CPC** was achieved by **Carousel** (3.0715), followed closely by **Image** (3.0082).
- The **lowest average CPC** was observed for **Display** (1.8540).
- Carousel and Image exhibited statistically similar CPC values, while Display and Search also did not differ significantly. All other pairwise comparisons showed statistically significant differences.

B.3.5 Video Completion Rate vs CVR

A correlation analysis was conducted to investigate the relationship between Video Completion Rate and Conversion Rate(CVR).

Descriptive Statistics

Table B.23 summarizes the descriptive statistics of the two variables.

Table B.23: Descriptive statistics for Video Completion Rate and Conversion Rate.

Variable	Count	Mean	Std. Dev.	Min	25%	Median	75%	Max
Video Completion Rate	1207	0.5758	0.0922	0.3995	0.5076	0.5903	0.6466	0.7410
CVR	1207	0.0264	0.0079	0.0078	0.0211	0.0264	0.0314	0.0521

Correlation Analysis

Person's correlation analysis yielded a weak negative correlation between Video Completion Rate and Conversion Rate ($r = -0.2024$, $p < 0.001$). Likewise, Spearman's rank correlation also indicated a weak negative monotonic relationship($\rho = -0.1832$, $p < 0.001$).

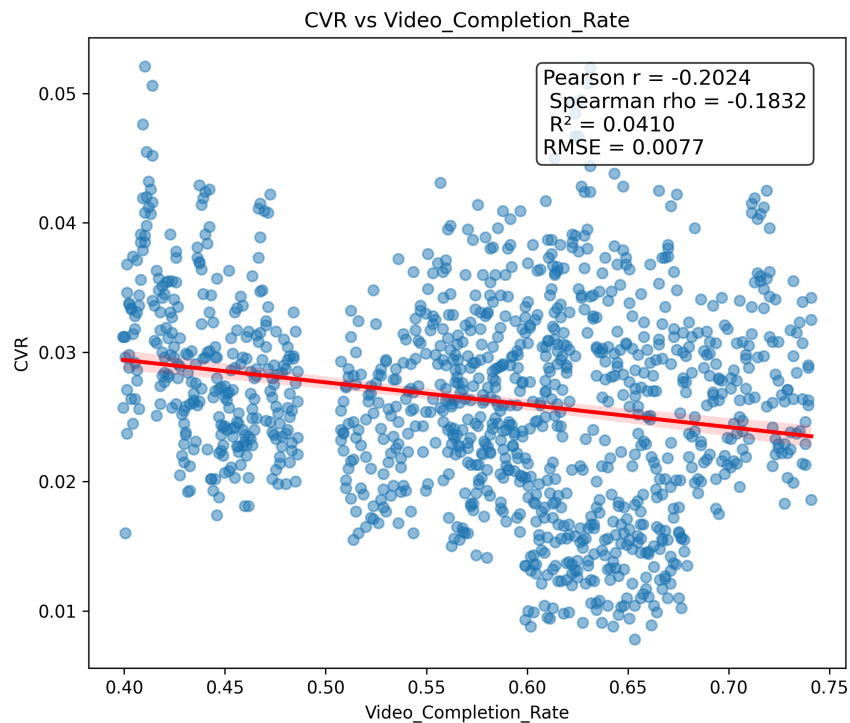


Figure B.1: CVR vs Video Completion Rate

These results suggests that campaigns with higher video completion rates tend to exhibit slightly lower conversion rates. However, the magnitude of the relationship is weak, indicating that Video Completion Rate alone is not a strong predictor of Conversion rate.

B.4 Relationship Between Video Completion Rate and Platform

B.4.1 Video Completion Rate by Platform

Descriptive Statistics

Table B.24: Descriptive statistics for Video Completion Rate by platform.

Platform	Mean	Median	Std. Dev.	Count
FB	0.4430	0.4442	0.0250	299
TT	0.6195	0.6163	0.0581	908

Assumption Tests**Shapiro–Wilk Test**

Platform	p-value
FB	< 0.0001
TT	< 0.0001

Both platform distributions significantly deviated from normality ($p < 0.05$).

Levene’s Test

Statistic	Value
p-value	< 0.0001

The variability of **Video Completion Rate** differed significantly between platforms. However, it is important to note that TT has 3 product_categories while FB has 1, so the difference in STD may be attributed to TT having more diverse product categories.

Kruskal–Wallis Test

Statistic	Value
H-statistic	674.2412
p-value	< 0.0001

The Kruskal–Wallis test indicates a statistically significant difference in Video Completion Rate between advertising platforms.

Dunn Post-hoc Test

Table B.25: Pairwise Dunn test p-values (Bonferroni adjusted).

	FB	TT
FB	–	< 0.0001
TT	< 0.0001	–

Facebook and TikTok exhibited statistically significant differences in Video Completion Rate, with TikTok achieving significantly higher completion rates than Facebook.

Summary

- The **highest average Video Completion Rate** was achieved by **TikTok** (0.6195).
- The **lowest average Video Completion Rate** was observed for **Facebook** (0.4430).
- **TikTok achieved significantly higher Video Completion Rates than Facebook** ($p < 0.05$).

B.4.2 Video Completion Rate by TT Platform by Product Category

Descriptive Statistics

Table B.26: Descriptive statistics for Video Completion Rate by product category.

Product Category	Mean	Median	Std. Dev.	Count
Diet	0.5618	0.5619	0.0320	293
Peworkout	0.6181	0.6181	0.0349	300
Protein	0.6745	0.6758	0.0387	315

Assumption Tests

Shapiro–Wilk Test

Product Category	p-value
Diet	< 0.0001
Peworkout	< 0.0001
Protein	< 0.0001

All product category distributions significantly deviated from normality ($p < 0.05$).

Levene’s Test

Statistic	Value
p-value	0.0001

The variability of **Video Completion Rate** differed significantly across product categories.

Kruskal–Wallis Test

Statistic	Value
H-statistic	587.1425
p-value	< 0.0001

The Kruskal–Wallis test indicates statistically significant differences in Video Completion Rate across product categories.

Dunn Post-hoc Test

Table B.27: Pairwise Dunn test p-values (Bonferroni adjusted).

	Diet	Peworkout	Protein
Diet	–	< 0.0001	< 0.0001
Peworkout	< 0.0001	–	< 0.0001
Protein	< 0.0001	< 0.0001	–

All pairwise comparisons between product categories were statistically significant, indicating that each product category achieved a distinct level of Video Completion Rate.

Within the TT Platform we have the following summary:

- The **highest average Video Completion Rate** was achieved by **Protein** (0.6745).
- The **lowest average Video Completion Rate** was observed for **Diet** (0.5618).
- **All product categories differed significantly** in Video Completion Rate ($p < 0.05$), with Protein achieving the highest completion rates, followed by Peworkout and Diet.

B.4.3 Video Completion Rate by Region

Descriptive Statistics

Table B.28: Descriptive statistics for Video Completion Rate by region.

Region	Mean	Median	Std. Dev.	Count
Midwest	0.6206	0.6188	0.0584	220
Northeast	0.6191	0.6165	0.0578	229
South	0.6199	0.6160	0.0587	228
West	0.6184	0.6140	0.0580	231

Assumption Tests

Shapiro–Wilk Test

Region	p-value
Midwest	0.0096
Northeast	0.0090
South	0.0028
West	0.0065

All regional distributions significantly deviated from normality ($p < 0.05$).

Levene's Test

Statistic	Value
p-value	0.9766

No statistically significant difference in the variability of **Video Completion Rate** was observed across regions.

Kruskal–Wallis Test

Statistic	Value
H-statistic	0.1893
p-value	0.9793

The Kruskal–Wallis test indicates that there are no statistically significant differences in Video Completion Rate across regions.

Summary

- The **highest average Video Completion Rate** was achieved by the **Midwest** (0.6206).
- The **lowest average Video Completion Rate** was observed for the **West** (0.6184).
- **No statistically significant differences** in Video Completion Rate were observed across regions ($p \geq 0.05$).

B.4.4 Video Completion Rate by Competitive Event

Descriptive Statistics

Table B.29: Descriptive statistics for Video Completion Rate by competitive event status.

Competitive Event	Mean	Median	Std. Dev.	Count
False	0.6189	0.6152	0.0580	872
True	0.6335	0.6195	0.0610	36

Assumption Tests

Shapiro–Wilk Test

Competitive Event	p-value
False	< 0.0001
True	0.0273

Both distributions significantly deviated from normality ($p < 0.05$).

Levene’s Test

Statistic	Value
p-value	0.9008

No statistically significant difference in the variability of **Video Completion Rate** was observed between competitive and non-competitive event periods.

Kruskal–Wallis Test

Statistic	Value
H-statistic	1.5983
p-value	0.2061

The Kruskal–Wallis test indicates that there are no statistically significant differences in Video Completion Rate between competitive and non-competitive event periods.

Summary

- The **highest average Video Completion Rate** was observed during **competitive events** (0.6335).
- The **lowest average Video Completion Rate** was observed during **non-competitive event periods** (0.6189).
- **No statistically significant difference** in Video Completion Rate was observed between competitive and non-competitive event periods ($p \geq 0.05$).

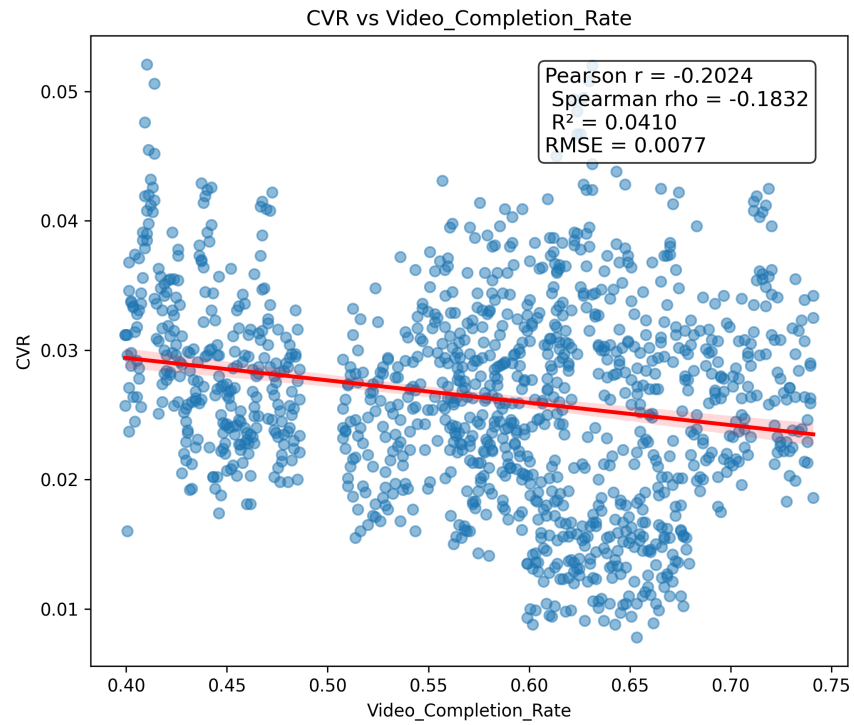


Figure B.2: CVR vs Video Completion Rate

B.4.5 Summary for Video Completion Rate by Platform

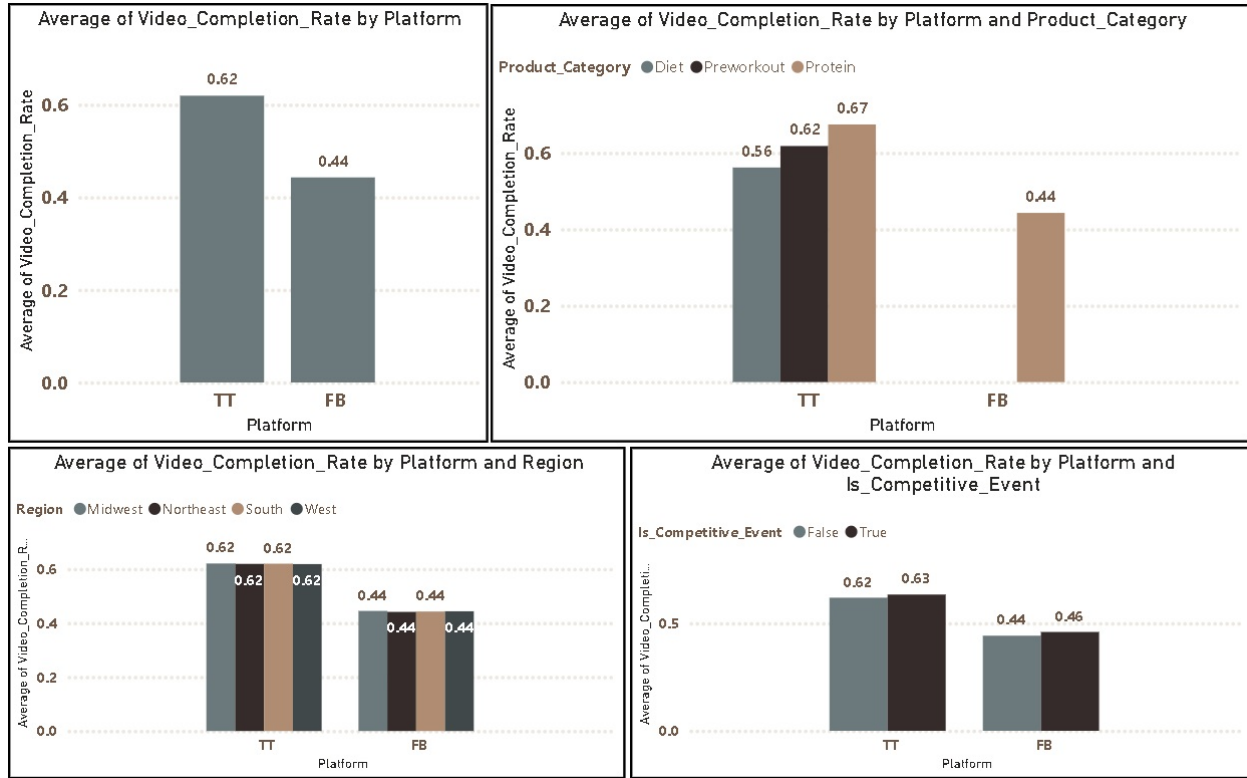


Figure B.3: Scatter plots of CVR versus Video Completion Rate grouped by platform, product category, competitive event status, and region

- Video ads from TT platform exhibited higher video completion rate for Protein product category [Appendix B.4.1].
- Within the TT platform, product category seem to significantly affect video completion rate where Protein product exhibits the highest video completion rate [Appendix B.4.2].
- Further tests, like FB video ads with Peworkout and Diet product categories are needed for evidence of Platform effects to video completion rate.
- Competitive events and Region seem to not affect the video completion rate across TT and FB Platforms.

Overall, product categories and platform(potentially) significantly affect video completion rate. These insights will be helpful for campaigns with a goal of maximizing video completion rate.

B.5 Audience-Product Performance Analysis

B.5.1 Return on Advertising Spend (ROAS)

Descriptive Statistics

Table B.30: Descriptive statistics for ROAS by audience-product combination.

Audience-Product	Mean	Median	Std. Dev.	Count
Athletes-Preworkout	0.6176	0.5656	0.2749	349
Athletes-Protein	1.3059	1.1981	0.5542	1054
Fitness Enthusiasts-Preworkout	0.6189	0.5754	0.2404	704
Fitness Enthusiasts-Weight Loss	0.7560	0.6572	0.4480	355
Weight Loss-Diet	1.0578	0.9574	0.5374	711

Assumption Tests

Shapiro-Wilk Test

Audience-Product	p-value
Athletes-Preworkout	< 0.0001
Athletes-Protein	< 0.0001
Fitness Enthusiasts-Preworkout	< 0.0001
Fitness Enthusiasts-Weight Loss	< 0.0001
Weight Loss-Diet	< 0.0001

All audience-product distributions significantly deviated from normality ($p < 0.05$).

Levene’s Test

Statistic	Value
p-value	< 0.0001

The variability of **ROAS differed significantly** across audience-product combinations.

Kruskal-Wallis Test

Statistic	Value
H-statistic	1089.1785
p-value	< 0.0001

The Kruskal–Wallis test indicates statistically significant differences in ROAS across audience–product combinations.

Dunn Post-hoc Test

Table B.31: Pairwise Dunn test p-values (Bonferroni adjusted).

	Ath-Pre	Ath-Pro	Fit-Pre	Fit-WL	WL-Diet
Ath-Pre	–	< 0.0001	1.0000	0.0008	< 0.0001
Ath-Pro	< 0.0001	–	< 0.0001	< 0.0001	< 0.0001
Fit-Pre	1.0000	< 0.0001	–	0.0001	< 0.0001
Fit-WL	0.0008	< 0.0001	0.0001	–	< 0.0001
WL-Diet	< 0.0001	< 0.0001	< 0.0001	< 0.0001	–

Summary

- The highest average ROAS was achieved by **Athletes–Protein** (1.3059).
- The lowest average ROAS was observed for **Athletes–Preworkout** (0.6176), although it did not differ significantly from **Fitness Enthusiasts–Preworkout**.
- **Athletes–Protein** achieved significantly higher ROAS than every other audience–product combination.
- Both Pre-workout audience pairings exhibited statistically equivalent ROAS, suggesting consistently poor profitability regardless of whether the campaign targeted athletes or fitness enthusiasts.

B.5.2 Click-Through Rate (CTR)

Descriptive Statistics

Table B.32: Descriptive statistics for CTR by audience–product combination.

Audience–Product	Mean	Median	Std. Dev.	Count
Athletes–Preworkout	0.0055	0.0052	0.0022	349
Athletes–Protein	0.0074	0.0067	0.0037	1054
Fitness Enthusiasts–Preworkout	0.0081	0.0071	0.0044	704
Fitness Enthusiasts–Weight Loss	0.0054	0.0051	0.0021	355
Weight Loss–Diet	0.0083	0.0073	0.0042	711

Assumption Tests**Shapiro-Wilk Test**

Audience-Product	p-value
Athletes-Preworkout	< 0.0001
Athletes-Protein	< 0.0001
Fitness Enthusiasts-Preworkout	< 0.0001
Fitness Enthusiasts-Weight Loss	< 0.0001
Weight Loss-Diet	< 0.0001

All audience-product distributions significantly deviated from normality ($p < 0.05$).

Levene's Test

Statistic	Value
p-value	< 0.0001

The variability of **CTR differed significantly** across audience-product combinations.

Kruskal-Wallis Test

Statistic	Value
H-statistic	281.8890
p-value	< 0.0001

The Kruskal-Wallis test indicates statistically significant differences in CTR across audience-product combinations.

Dunn Post-hoc Test

Table B.33: Pairwise Dunn test p-values (Bonferroni adjusted).

	Ath-Pre	Ath-Pro	Fit-Pre	Fit-WL	WL-Diet
Ath-Pre	–	< 0.0001	< 0.0001	1.0000	< 0.0001
Ath-Pro	< 0.0001	–	0.0284	< 0.0001	0.0002
Fit-Pre	< 0.0001	0.0284	–	< 0.0001	1.0000
Fit-WL	1.0000	< 0.0001	< 0.0001	–	< 0.0001
WL-Diet	< 0.0001	0.0002	1.0000	< 0.0001	–

Summary

- The highest average CTR was achieved by **Weight Loss–Diet** (0.0083).
- The lowest average CTR was observed for **Fitness Enthusiasts–Weight Loss** (0.0054).
- **Weight Loss–Diet** and **Fitness Enthusiasts–Preworkout** exhibited statistically equivalent CTR values.
- Likewise, **Athletes–Preworkout** and **Fitness Enthusiasts–Weight Loss** showed no statistically significant difference in CTR, whereas all remaining pairwise comparisons were statistically significant.

B.5.3 Conversion Rate (CVR)

Descriptive Statistics

Table B.34: Descriptive statistics for CVR by audience–product combination.

Audience–Product	Mean	Median	Std. Dev.	Count
Athletes–Preworkout	0.0265	0.0252	0.0081	349
Athletes–Protein	0.0300	0.0293	0.0066	1017
Fitness Enthusiasts–Preworkout	0.0179	0.0169	0.0055	666
Fitness Enthusiasts–Weight Loss	0.0300	0.0289	0.0078	355
Weight Loss–Diet	0.0280	0.0278	0.0059	687

Assumption Tests

Shapiro–Wilk Test

Audience–Product	p-value
Athletes–Preworkout	< 0.0001
Athletes–Protein	< 0.0001
Fitness Enthusiasts–Preworkout	< 0.0001
Fitness Enthusiasts–Weight Loss	< 0.0001
Weight Loss–Diet	0.0050

All audience–product distributions significantly deviated from normality ($p < 0.05$).

Levene’s Test

Statistic	Value
p-value	< 0.0001

The variability of **CVR** differed significantly across audience–product combinations.

Kruskal-Wallis Test

Statistic	Value
H-statistic	1058.9655
p-value	< 0.0001

The Kruskal-Wallis test indicates statistically significant differences in CVR across audience-product combinations.

Dunn Post-hoc Test

Table B.35: Pairwise Dunn test p-values (Bonferroni adjusted).

	Ath-Pre	Ath-Pro	Fit-Pre	Fit-WL	WL-Diet
Ath-Pre	–	< 0.0001	< 0.0001	< 0.0001	0.0005
Ath-Pro	< 0.0001	–	< 0.0001	1.0000	0.0002
Fit-Pre	< 0.0001	< 0.0001	–	< 0.0001	< 0.0001
Fit-WL	< 0.0001	1.0000	< 0.0001	–	0.0802
WL-Diet	0.0005	0.0002	< 0.0001	0.0802	–

Summary

- The highest average CVR was achieved jointly by **Athletes-Protein** and **Fitness Enthusiasts-Weight Loss** (0.0300).
- The lowest average CVR was observed for **Fitness Enthusiasts-Preworkout** (0.0179).
- **Athletes-Protein** and **Fitness Enthusiasts-Weight Loss** exhibited statistically equivalent conversion rates.
- Likewise, **Fitness Enthusiasts-Weight Loss** and **Weight Loss-Diet** did not differ significantly in CVR, whereas all remaining pairwise comparisons were statistically significant.

B.5.4 Cost Per Click (CPC)**Descriptive Statistics**

Table B.36: Descriptive statistics for CPC by audience-product combination.

Audience-Product	Mean	Median	Std. Dev.	Count
Athletes-Preworkout	3.0082	2.6030	1.4140	349
Athletes-Protein	2.1128	1.9627	0.8525	1017
Fitness Enthusiasts-Preworkout	1.9238	1.7504	0.7849	666
Fitness Enthusiasts-Weight Loss	3.0715	2.6470	1.4731	355
Weight Loss-Diet	1.9270	1.7633	0.7977	687

Assumption Tests**Shapiro–Wilk Test**

Audience–Product	p-value
Athletes–Preworkout	< 0.0001
Athletes–Protein	< 0.0001
Fitness Enthusiasts–Preworkout	< 0.0001
Fitness Enthusiasts–Weight Loss	< 0.0001
Weight Loss–Diet	< 0.0001

All audience–product distributions significantly deviated from normality ($p < 0.05$).

Levene’s Test

Statistic	Value
p-value	< 0.0001

The variability of **CPC differed significantly** across audience–product combinations.

Kruskal–Wallis Test

Statistic	Value
H-statistic	368.5351
p-value	< 0.0001

The Kruskal–Wallis test indicates statistically significant differences in CPC across audience–product combinations.

Dunn Post-hoc Test

Table B.37: Pairwise Dunn test p-values (Bonferroni adjusted).

	Ath-Pre	Ath-Pro	Fit-Pre	Fit-WL	WL-Diet
Ath-Pre	–	< 0.0001	< 0.0001	1.0000	< 0.0001
Ath-Pro	< 0.0001	–	0.0002	< 0.0001	0.0002
Fit-Pre	< 0.0001	0.0002	–	< 0.0001	1.0000
Fit-WL	1.0000	< 0.0001	< 0.0001	–	< 0.0001
WL-Diet	< 0.0001	0.0002	1.0000	< 0.0001	–

Summary

- The highest average CPC was achieved by **Fitness Enthusiasts–Weight Loss** (3.0715).
- The lowest average CPC was observed for **Fitness Enthusiasts–Preworkout** (1.9238), although it did not differ significantly from **Weight Loss–Diet**.
- **Athletes–Preworkout** and **Fitness Enthusiasts–Weight Loss** exhibited statistically equivalent CPC values.
- Likewise, **Fitness Enthusiasts–Preworkout** and **Weight Loss–Diet** showed no statistically significant difference in CPC, whereas all remaining pairwise comparisons were statistically significant.

B.6 Statistical Test Results of Competitive Event Analysis

This appendix presents the statistical test results evaluating whether campaign performance differed between periods with and without competitive events.

B.6.1 Return on Advertising Spend (ROAS)

Descriptive Statistics

Table B.38: Descriptive statistics for ROAS by competitive event.

Competitive Event	Mean	Median	Std. Dev.	Count
False	0.9664	0.8475	0.5460	3031
True	0.8383	0.7503	0.4729	142

Statistical Tests

Test	p-value
Shapiro–Wilk (False)	< 0.0001
Shapiro–Wilk (True)	< 0.0001
Levene’s Test	0.0276
Kruskal–Wallis Test	0.0038

Campaigns conducted during competitive events exhibited significantly lower ROAS than campaigns conducted during non-competitive periods.

B.6.2 Click-Through Rate (CTR)

Descriptive Statistics

Table B.39: Descriptive statistics for CTR by competitive event.

Competitive Event	Mean	Median	Std. Dev.	Count
False	0.0073	0.0065	0.0038	3031
True	0.0088	0.0083	0.0045	142

Statistical Tests

Test	p-value
Shapiro–Wilk (False)	< 0.0001
Shapiro–Wilk (True)	< 0.0001
Levene’s Test	0.0772
Kruskal–Wallis Test	< 0.0001

Campaigns conducted during competitive events achieved significantly higher CTR than campaigns conducted during non-competitive periods.

B.6.3 Conversion Rate (CVR)

Descriptive Statistics

Table B.40: Descriptive statistics for CVR by competitive event.

Competitive Event	Mean	Median	Std. Dev.	Count
False	0.0265	0.0264	0.0081	2936
True	0.0265	0.0264	0.0082	138

Statistical Tests

Test	p-value
Shapiro–Wilk (False)	< 0.0001
Shapiro–Wilk (True)	0.6173
Levene’s Test	0.7197
Kruskal–Wallis Test	0.9415

No statistically significant difference in CVR was observed between competitive and non-competitive periods.

B.6.4 Cost Per Click (CPC)

Descriptive Statistics

Table B.41: Descriptive statistics for CPC by competitive event.

Competitive Event	Mean	Median	Std. Dev.	Count
False	2.2335	1.9756	1.0956	2936
True	2.4392	2.2751	0.9424	138

Statistical Tests

Test	p-value
Shapiro–Wilk (False)	< 0.0001
Shapiro–Wilk (True)	0.0001
Levene’s Test	0.5691
Kruskal–Wallis Test	0.0006

Campaigns conducted during competitive events incurred significantly higher CPC than campaigns conducted during non-competitive periods.

B.6.5 Summary

- Competitive events were associated with a statistically significant decrease in ROAS.
- Campaigns conducted during competitive events generated significantly higher CTR.
- No statistically significant difference in CVR was observed between competitive and non-competitive periods.
- CPC increased significantly during competitive events, indicating greater advertising costs to acquire user clicks.
- Collectively, these findings suggest that reduced profitability during competitive events is primarily driven by higher acquisition costs rather than changes in customers’ likelihood of purchasing after clicking an advertisement.